COMPARATIVE ANALYSIS OF LARGE LANGUAGE MODELS (LLMS)

For Retrieval-Augmented Generation (RAG)

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PREPARED BY :

Lashawn Fofung, MBA,PMP, CSM, CSPO

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TECHNICAL EVALUATION

LLMs Speed and Accuracy for RAG

Date of Analysis: 12/11/2025

Objective: Systematically evaluate the speed and factual accuracy of proprietary (Gemini) and open-source (Mistral 7B, Phi-2, TinyLlama) LLMs within a Retrieval-Augmented Generation (RAG) pipeline to inform model selection for domain-specific query systems.

GitHub:
https://github.com/LashawnFofung/RAG-Pipelines/blob/main/src/Task_LLM_Evaluation_for_RAG.ipynb

EXECUTIVE SUMMARY

This evaluation focused on benchmarking four distinct Large Language Models (LLMs) within a RAG pipeline against a sensitive, domain-specific document (e.g., mortgage contract). The optimal model choice depends on a critical trade-off between factual accuracy (best served by larger models) and inference speed/cost efficiency (best served by smaller, highly optimized models).

- **Best Factual Accuracy (Qualitative Winner):** Gemini demonstrated superior reasoning and consistently high performance, making it the professional baseline for accuracy and complex reasoning tasks required in regulated industries.
- **Fastest Inference (Quantitative Winner):** Mistral 7B (GGUF) was the most efficient model tested locally, achieving the fastest average query speed of 3.81 seconds (Average Speed: 3.30s in summarization task).
- **Best Value Proposition:** Mistral 7B offers an excellent blend of generation quality, low latency, and competitive cost, making it ideal for general-purpose high-throughput RAG systems.

CAPABILITY

The core limitation of traditional LLMs is their reliance on static training data. RAG addresses this by augmenting the LLM prompt with external, verifiable documents, enhancing both the factual accuracy (Faithfulness) and timeliness of the response. The models below were evaluated for their capability to utilize retrieved context to generate accurate and relevant answers.

Feature	Developer	Model Type	RAG Role
Gemini (External API)	Google	Closed/ Proprietary	High-precision, complex reasoning, handling very long contexts.
Mistral 7B (GGUF)	Mistral AI	Open-Weight (Commercial use)	Balanced performance/cost, excellent for high-throughput and low-latency RAG.
Phi-2 (Microsoft)	Microsoft	Open-Weight (Commercial use)	Edge deployment, highly cost-efficient, simple Q&A RAG.
TinyLlama (1.1B)	Open-Source Community	Open-Source (Smallest Footprint)	Ultimate test for fast, resource-constrained environments.

QUANTITATIVE PERFORMANCE METRICS

The models were tested using a single, uniform query ("What are the penalties for late payments?") within a LlamaIndex RAG engine, comparing an external API model against locally run, on-GPU open-source models.

Model	Average Query Speed (s)	Response (Excerpt)	Total Queries
Mistral 7B	3.81	1.5% per month interest will be charged on late payments until they are paid in full.....	2
TinyLlama	5.86	1.5% per month from the due date until paid in full.....	2
Gemini	9.41	Late payments will incur interest at a rate of 1.5% per month, calculated from the due date until th...	2
Phi-2	10.33	The late fee is fifty dollars ($50) if received after the 5th day of the month.....	2

Key Finding:

- **Speed Discrepancy:** The open-source, optimized Mistral 7B was significantly faster than both the external Gemini API and the other Small Language Models (SLMs) in this specific RAG setup.
- **Phi-2 and TinyLlama Performance:** Despite their smaller parameter size, both Phi-2 and TinyLlama showed higher latency in this specific measured test (10.33s and 5.86s, respectively) than the Mistral GGUF configuration.

QUALITATIVE RETRIEVAL SCORECARD

The quality of the RAG output is measured by the LLM's ability to remain Faithful to the retrieved context and generate a Relevant answer.

Model	Faithfulness	Answer Relevance	Example Accuracy
Focus	Generation Quality	Generation Quality	Grounding
Definition	Proportion of facts in the answer supported by retrieved context.	Assesses if the generated answer directly addresses the user's query.	Comparison of the example answer to a factual source.
Gemini	Highest Potential: Accurately synthesizes information from complex contexts, sticking strictly to provided facts.	Highest Quality RAG: Low latency for complex summarization, showing high RAG efficiency.	Accurate: 1.5% interest rate.
Mistral 7B	Strong Performance: Delivers reliable output, striking a balance between context adherence and coherence.	Robust Generator: Reliable instruction following to produce coherent and relevant answers.	Accurate: 1.5% interest rate.
Phi-2 / TinyLlama	Lower Potential: May struggle with long or conflicting contexts, occasionally introducing subtle inaccuracies.	Highly dependent on retrieved context quality; less capable of complex contextual re-ranking.	Inaccurate/Misinterpreted:$50 late fee (contradicts 1.5% interest in other models).

Key Finding:

- **Factual Fidelity:** Gemini and Mistral 7B provided consistent answers regarding the 1.5% interest penalty, suggesting stronger factual grounding. Phi-2 provided a conflicting response of a "$50 late fee," indicating a potential failure in synthesizing the correct information from the retrieved context or a hallucination.

COMPARATIVE STRENGTHS & OPERATIONAL BENEFIT

Model	Key Strength	Operational Benefit
🧠 **Gemini**	State-of-the-art reasoning and multimodal capabilities.	Top choice for use cases requiring maximum factual accuracy (e.g., legal, medical RAG).
🧠 **Mistral 7B**	Exceptional efficiency and competitive performance for its size.	Strong contender for mid-to-high-throughput enterprise RAG due to high speed and low cost.
🧠 **Phi-2/TinyLlama**	Lowest parameter count (SLMs), low resource cost.	Ideal for high-volume, low-latency applications or edge deployment where speed is paramount.

Trade-off / Consideration:

- **Gemini:** Highest latency and operational cost.
- **Mistral 7B:** Performance may be surpassed by Gemini on highly complex, unstructured documents.
- **Phi-2/TinyLlama:** High reliance on a perfect retrieval system; greater risk of missing nuance or introducing inaccuracies in complex RAG.

CONCLUSION

The optimal LLM for a RAG system is a decision based purely on balancing the non-negotiable requirements of the project (accuracy vs. speed vs. cost).

Use Case	Recommended Model / Strategy	Rationale
High-Accuracy / Complex Reasoning	🧠 **Gemini**	Superior reasoning and factual synthesis from complex, long contexts. Best for regulated industries or critical decision-support systems.
Balanced Performance & Throughput	🧠 **Mistral 7B**	Excellent blend of generation quality and low latency (3.81s measured speed). Ideal for general-purpose customer service or internal knowledge retrieval systems.
Cost-Sensitive/ Edge Deployment	🧠 **Phi-2/TinyLlama**	Lowest inference cost (theoretically) and smallest memory footprint. Perfect for mobile or local, high-volume Q&A where the knowledge base is simple.

The comparative evaluation confirms a crucial trade-off in the RAG pipeline: the balance between maximal factual accuracy and system performance. The proprietary Gemini model establishes the gold standard for high-fidelity reasoning and context synthesis, making it the superior choice for high-stakes, regulated environments where minimizing hallucination is paramount. Conversely, the optimized, open-source Mistral 7B (GGUF) emerged as the champion for efficiency, delivering the fastest average inference speed (3.81 seconds) and making it the ideal selection for high-throughput, general-purpose RAG where a balance of quality and speed is necessary. Models like Phi-2 and TinyLlama, while offering the lowest operational cost, present a greater risk of factual inconsistency, demonstrating that model selection must be strategically aligned with the organization's specific risk profile and operational needs.

PROJECT & PLATFORM SUMMARY

AI-powered Document Automation Platform Project

This analysis was conducted as part of the rigorous testing phase for the AI-powered Document Automation Platform. The platform's primary goal is to provide clients with sub-second retrieval from complex, large document files, thereby guaranteeing both speed and factual accuracy.

The comparative evaluation demonstrates the platform's commitment to evidence-based model selection. The Large Language Model (LLM) component was benchmarked, revealing a strategic choice between the high-fidelity reasoning of the Gemini model and the superior speed and balanced quality of the optimized, open-source Mistral 7B (GGUF). This comprehensive LLM testing ensures the final architecture is tuned to balance maximal accuracy for sensitive queries against high throughput for scalable performance, ultimately transforming complex data into reliable, actionable intelligence

You can learn more about the project by visiting:

- GitHub: https://github.com/LashawnFofung/AI-Powered-Document-Automation-Platform

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